

Deep Learning Solvers as Lamphrodynamic Flows in RSVP Field Theory

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Abstract

Deep learning methods for solving high-dimensional dynamic economic models have recently gained attention for their ability to “unbend” complex solution manifolds and overcome the curse of dimensionality. This essay presents a unified geometric and variational interpretation of Jess Fernández-Villaverdes deep learning solvers through the lens of the Relativistic Scalar Vector Plenum (RSVP) field theory. By identifying value functions, policy rules, and informational structures with scalar potentials, vector flows, and entropy fields in RSVP, we show that deep networks implement a lamphrodynamic smoothing flow that discovers the natural coordinate charts of the economic equilibrium manifold. We derive a free-energy functional whose minimization is equivalent to satisfying equilibrium conditions, formalize the associated gradient flow, and prove equivalence theorems linking economic equilibrium with RSVP stationary field configurations.

1 Introduction

In recent work, Fernández-Villaverde has argued that deep neural networks provide a powerful and flexible method for solving dynamic economic models, especially in high-dimensional settings where traditional numerical approaches become intractable. He emphasizes the geometric role of deep networks as machines that “unbend” complex high-dimensional solution manifolds into simpler latent representations.

This geometric perspective aligns naturally with the Relativistic Scalar Vector Plenum (RSVP), a field-theoretic framework in which economic objects are embedded as scalar, vector, and entropy fields living on a curved manifold. RSVP provides an intrinsic geometric and variational interpretation of deep learning solvers as *lamphrodynamic flows*—curvature-smoothing, entropy-reducing transformations that evolve field configurations toward equilibrium.

This essay develops the full mapping between the economic, deep learning, and RSVP perspectives, culminating in a free-energy functional whose minimization yields an economic equilibrium.

2 Equilibrium as a Field Configuration

Dynamic economic models require finding a function

$$D(\mathbf{x}) \tag{1}$$

that satisfies an equilibrium operator equation:

$$T(D) = \mathbf{0}, \tag{2}$$

where D may represent a value function, policy rule, or both.

RSVP Field Mapping

In RSVP, the unknown function D corresponds to a tuple of fields:

$$(\Phi, \mathbf{v}, S),$$

where

- $\Phi(\mathbf{x})$: a scalar potential (value function),
- $\mathbf{v}(\mathbf{x})$: a vector flow (policy dynamics),
- $S(\mathbf{x})$: an entropy field (uncertainty, beliefs).

Thus the equilibrium becomes a geometric constraint:

$$T(\Phi, \mathbf{v}, S) = \mathbf{0}. \tag{3}$$

Definition 1 (RSVP Equilibrium Field). *A triple $(\Phi, \mathbf{v}, S)$ is an RSVP equilibrium if it satisfies*

$$T(\Phi, \mathbf{v}, S) = 0.$$

This is analogous to a fixed point, zero-curvature state, or a stationary point of a variational energy.

3 Dimensionality as Curvature and Entropy

The curse of dimensionality arises when the solution surface is highly curved or kinked. RSVP identifies:

- **Curvature:** large $\|\nabla\Phi\|$ or $\|\nabla\mathbf{v}\|$,
- **Torsion:** nondifferentiabilities induced by constraints,
- **Entropy-of-representation:** irregular information distribution in S .

Standard bases (e.g., polynomials) assume low curvature and low entropy. Deep networks, by contrast, learn transformations that reduce both.

4 Neural Networks as Lamphrodynamic Operators

A single neural layer performs the map

$$f_{\theta}(\mathbf{x}) = \sigma(W\mathbf{x} + b), \quad (4)$$

a local nonlinear transformation smoothing curvature and concentrating entropy. RSVP identifies such a transformation as a *lamphrodyne update*.

A deep network composed of L such layers:

$$\mathbf{x} \xrightarrow{f_1} \mathbf{x}_1 \xrightarrow{f_2} \dots \xrightarrow{f_L} \mathbf{x}_L$$

approximates a continuous lamphrodynamic flow:

$$\frac{d\mathbf{x}}{dt} = \mathcal{V}(\mathbf{x}), \quad (5)$$

where $\mathcal{V}$ is a curvature-smoothing vector field.

5 A Free-Energy Functional for Economic Equilibrium

To interpret training as variational descent, we introduce a free-energy functional.

Definition 2 (RSVP Free-Energy Functional). *Let $D = D(\Phi, \mathbf{v}, S)$. Define:*

$$\mathcal{E}[D] = \frac{1}{2} \int_{\mathcal{X}} \|T(D(\mathbf{x}))\|^2 d\mu(\mathbf{x}). \quad (6)$$

We now prove that minimizing this energy is equivalent to satisfying the equilibrium operator.

Lemma 1 (First Variation of $\mathcal{E}$). *Let $D_\epsilon = D + \epsilon\delta D$. Then*

$$\left. \frac{d}{d\epsilon} \mathcal{E}[D_\epsilon] \right|_{\epsilon=0} = \int_{\mathcal{X}} T(D) \frac{\delta T}{\delta D}[\delta D] d\mu.$$

Proof. Expand:

$$\mathcal{E}[D_\epsilon] = \frac{1}{2} \int \|T(D + \epsilon\delta D)\|^2 d\mu.$$

Differentiate under the integral:

$$\frac{d}{d\epsilon} \mathcal{E}[D_\epsilon] = \int T(D) \cdot \frac{d}{d\epsilon} T(D + \epsilon\delta D) d\mu.$$

At $\epsilon = 0$, this yields the stated expression. □

Theorem 1 (Equivalence of Minimizers and Equilibria). *A field configuration D^* minimizes $\mathcal{E}$ if and only if*

$$T(D^*) = 0.$$

Proof. If $T(D^*) = 0$, then $\mathcal{E}[D^*] = 0$ and is clearly minimal.

Conversely, suppose D^* minimizes $\mathcal{E}$. Then the first variation must vanish for all perturbations:

$$\int T(D^*) \frac{\delta T}{\delta D}[\delta D] d\mu = 0 \quad \forall \delta D.$$

If $\delta T/\delta D$ is nondegenerate, this implies $T(D^*) = 0$. □

Proposition 1. *Gradient descent on $\mathcal{E}$ generates a lamphrodynamical flow*

$$\begin{aligned} \frac{d\Phi}{dt} &= -\frac{\delta \mathcal{E}}{\delta \Phi}, \\ \frac{d\mathbf{v}}{dt} &= -\frac{\delta \mathcal{E}}{\delta \mathbf{v}}, \\ \frac{dS}{dt} &= -\frac{\delta \mathcal{E}}{\delta S}. \end{aligned} \tag{7}$$

Proof. Immediate from functional gradient descent applied to $\mathcal{E}[\Phi, \mathbf{v}, S]$. □

6 Formal RSVP Formulation of Dynamic Economic Models

Let $\mathcal{X}$ be the state manifold. Define:

$$\Phi : \mathcal{X} \rightarrow \mathbb{R}, \quad \mathbf{v} : \mathcal{X} \rightarrow T\mathcal{X}, \quad S : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}.$$

Definition 3 (RSVP Economic Model). *A dynamic economic model is the tuple $(\mathcal{X}, \Phi, \mathbf{v}, S, T)$ satisfying*

$$T(\Phi, \mathbf{v}, S) = 0.$$

Theorem 2 (RSVP Characterization of Equilibrium). *The economic equilibrium is the stationary point of the lamphrodynamic gradient flow on the free-energy functional:*

$$(\Phi^*, \mathbf{v}^*, S^*) = \arg \min_{(\Phi, \mathbf{v}, S)} \mathcal{E}[\Phi, \mathbf{v}, S].$$

Proof. Direct from the previous theorem and definitions. □

7 Variational and Field-Theoretic Perspectives

The lamphrodynamic interpretation of deep learning as a curvature-reducing flow naturally invites a broader variational and field-theoretic perspective. Within this conceptual frame, an economic equilibrium is not merely the fixed point of an operator, but the stationary configuration of a collection of fields whose interactions encode incentives, information, constraints, and adaptive dynamics.

In the standard formulation, one seeks a function $D(\mathbf{x})$ —a value function, policy function, or other decision rule—that satisfies an operator equation $T(D) = 0$. In the geometric formulation adopted here, the same object is represented by a triplet of fields

$$(\Phi, \mathbf{v}, S),$$

where Φ denotes a scalar potential associated with accumulated value or fitness, $\mathbf{v}$ denotes a vector flow describing the directional structure of optimal adjustment, and S denotes an entropy or informational field induced by beliefs, dispersion, or uncertainty.

From this vantage, the equilibrium condition $T(D) = 0$ admits a natural translation into a stationarity requirement for a free-energy-like functional $\mathcal{E}[\Phi, \mathbf{v}, S]$. The discrete steps of a numerical method, including those implemented by deep learning procedures, can then be viewed as approximations to a continuous relaxation dynamics in which the fields evolve toward the stationary configuration:

$$\frac{\delta \mathcal{E}}{\delta \Phi} = \frac{\delta \mathcal{E}}{\delta \mathbf{v}} = \frac{\delta \mathcal{E}}{\delta S} = 0.$$

This variational interpretation does not replace the computational perspective; instead, it clarifies why certain numerical schemes succeed, why they produce stable solutions, and why they behave as if they were performing implicit geometric simplifications of the underlying manifold. The economic equilibrium emerges, in this light, as the minimal-action configuration of an interacting field system.

8 Dynamics, Canonical Structure, and Information Flow

Once the fields $(\Phi, \mathbf{v}, S)$ are treated as dynamical quantities, it is natural to inquire whether the relaxation toward equilibrium admits a canonical structure. The temporal evolution of these fields defines an infinite dimensional phase space, in which each field possesses a conjugate momentum derived from its role in the Lagrangian or free-energy functional.

In this representation, iterative learning procedures, policy iterations, stochastic updates, and gradient-based optimization can all be interpreted as discrete approximations to continuous flows generated by a Hamiltonian or generalized Hamilton–Jacobi structure. The vector field governing these flows is not arbitrary: it encodes the propagation of information, the dissipation of irregularity, and the alignment of beliefs and actions, in exactly the way that canonical flows transmit forces, curvature, or probability mass in classical and statistical field theories.

This canonical viewpoint distinguishes genuine economic adjustment directions from redundant or gauge-like transformations of the fields. In particular, it reveals the extent to which variations in $\mathbf{v}$ reflect actual behavioral or decision-theoretic changes, while variations in S reflect informational or representational pressures. The interplay between these components can then be treated as a structured flow that respects both the geometry of the state manifold and the integrability constraints implied by economic rationality.

This dynamical framework provides a unifying structure in which learning, adaptation, inference, and optimization appear as different manifestations of a single, coherent geometric process.

9 Scale Interactions and Global Inference

Economic systems exhibit interactions across multiple scales: individual decisions respond to local conditions and expectations, while aggregate outcomes depend on global distributions and long-range informational structure. The field-theoretic representation makes these scale interactions explicit by encoding them in the triplet $(\Phi, \mathbf{v}, S)$.

The scalar potential Φ captures smooth, low-frequency structure such as persistent value gradients or intertemporal incentives. The vector field $\mathbf{v}$ describes local adjustment dynamics and short-scale sensitivity. The entropy field S records global dispersion, uncertainty, heterogeneity, and belief formation. Together, these fields determine how local optimization criteria propagate, how global consistency is enforced, and how equilibrium configurations aggregate information.

This structure admits a natural interpretation in terms of global inference problems. High-dimensional decision environments often contain latent low-dimensional order, but this structure may not be visible in the raw state space. Deep learning solvers implicitly identify this order by performing multi-scale smoothing and alignment steps, which correspond

to reducing the curvature and entropy of the fields while preserving the essential causal relationships.

From this perspective, the success of deep learning methods in solving high-dimensional equilibrium models is not accidental. These methods perform a sequence of transformations that resemble renormalization or coarse-graining: they reduce representational complexity while retaining the structures relevant for equilibrium. The final solution is therefore best understood as the fixed point of a multi-scale information flow, in which local incentives and global constraints coexist coherently.

This multi-scale geometric perspective offers new tools for analyzing stability, bifurcations, aggregation, and robustness in large economic systems, while preserving the interpretability and normative structure of equilibrium analysis.

10 Equilibrium, Information, and Futarchic Governance

If the lamphrodynamic interpretation of deep learning reveals the intrinsic geometry of equilibrium, it also clarifies the informational structure that any governance system must confront. In particular, the RSVP formulation offers a natural bridge to Hansons futarchy paradigm, in which policy selection is guided by predictive markets whose prices encode beliefs about welfare-relevant futures.

In this setting, the fields $(\Phi, \mathbf{v}, S)$ provide a precise analogue of the informational, behavioral, and uncertainty components that futarchy seeks to aggregate. The scalar potential Φ represents an intertemporal welfare or value landscape; the vector field $\mathbf{v}$ captures feasible adjustments and behavioral responses; and the entropy field S encodes the dispersion of beliefs, expectations, and informational frictions. Policy evaluation becomes a question of how proposed interventions deform these fields, alter the curvature of the welfare landscape, or redistribute entropy across states.

From this perspective, prediction markets do not merely forecast scalar aggregates; they express constraints on the allowable field configurations. Market-implied expectations correspond to boundary data for the fields, and policy comparison becomes the evaluation of alternative stationary solutions under different boundary or coupling structures. The futarchic principle “vote on values, bet on beliefs” thereby acquires a geometric interpretation: collective preferences determine the potential to be optimized, while market prices determine the informational terms that regulate how beliefs shape the flow of the fields.

This integration suggests that futarchy is not an institutional curiosity but a natural extension of a field-theoretic understanding of equilibrium. The same variational logic that governs economic dynamics, representation learning, and lamphrodynamic smoothing also governs the aggregation of dispersed information about the future. A futarchic mechanism can therefore be seen as selecting, from among many possible stationary field configurations, the one that maximizes collective welfare subject to informational coherence. This provides a

unified mathematical and conceptual foundation linking equilibrium computation, predictive governance, and high-dimensional learning dynamics.

11 A Formal Variational Theorem for Futarchic Policy Selection

The geometric interpretation of futarchy can be made precise by formulating policy choice as a constrained variational problem on the RSVP fields $(\Phi, \mathbf{v}, S)$. Let θ denote a feasible policy parameter, and let the resulting economic environment induce a free-energy functional $\mathcal{E}_\theta[\Phi, \mathbf{v}, S]$ whose stationary points correspond to equilibrium field configurations. Prediction markets generate belief-weighted boundary data that constrain the admissible fields. Let $\mathcal{B}_\theta$ denote this set of boundary conditions for policy θ .

We now formalize the relationship between futarchy and equilibrium selection.

Theorem 3 (Futarchic Stationarity Principle). *Suppose that for each feasible policy parameter θ , the free-energy functional $\mathcal{E}_\theta$ is convex in a neighborhood of its stationary points and that the associated boundary set $\mathcal{B}_\theta$ is nonempty, closed, and defined by market-implied expectations of welfare-relevant outcomes. Then the futarchic mechanism selects the policy θ^* whose equilibrium field configuration $(\Phi_\theta, \mathbf{v}_\theta, S_\theta)$ satisfies*

$$(\Phi_\theta, \mathbf{v}_\theta, S_\theta) \in_{(\Phi, \mathbf{v}, S) \in \mathcal{B}_\theta} \mathcal{E}_\theta[\Phi, \mathbf{v}, S],$$

and

$$\theta^* \in_\theta W[\Phi_\theta, \mathbf{v}_\theta, S_\theta],$$

where W is a welfare functional encoding the value-judgment component of collective preference. Moreover, θ^* is stable under perturbations of the boundary data corresponding to small revisions in market beliefs.

Proof. For any fixed θ , convexity of $\mathcal{E}_\theta$ in a neighborhood of its stationary points ensures the existence of a unique minimizer subject to $\mathcal{B}_\theta$. Since $\mathcal{B}_\theta$ consists of boundary data derived from prediction-market prices, the minimizer incorporates all belief-weighted expectations about future states.

Let $(\Phi_\theta, \mathbf{v}_\theta, S_\theta)$ denote this constrained minimizer. By definition, it satisfies the variational stationarity conditions

$$\frac{\delta \mathcal{E}_\theta}{\delta \Phi} = 0, \quad \frac{\delta \mathcal{E}_\theta}{\delta \mathbf{v}} = 0, \quad \frac{\delta \mathcal{E}_\theta}{\delta S} = 0 \quad \text{on the interior,}$$

together with the boundary constraints specified by $\mathcal{B}_\theta$.

Collective value judgments determine the welfare functional W . A futarchic mechanism chooses the policy that maximizes expected welfare evaluated at the equilibrium configuration. Thus,

$$\theta^* =_{\theta} W[\Phi_{\theta}, \mathbf{v}_{\theta}, S_{\theta}].$$

Finally, stability follows from the closedness of $\mathcal{B}_{\theta}$ and continuity of both the minimizer and the welfare functional in the underlying boundary data. Small perturbations in prediction-market prices induce small changes in the boundary set, which in turn induce small variations in the stationary solution and therefore do not alter the maximizing policy except under degeneracy. $\square$

This result shows that futarchy can be interpreted as a two-level variational problem: fields evolve internally to satisfy equilibrium constraints, while policy parameters are selected externally to maximize welfare under the belief-weighted informational structure encoded by markets. From this perspective, futarchy implements a principled mechanism for selecting among equilibrium field configurations under uncertainty, fully consistent with the variational and geometric structures underlying the RSVP representation.

Conclusion

Deep learning solvers for economic models naturally implement a form of lamphrodynamic flow in the sense of RSVP field theory. Neural networks act as automatic diffeomorphism machines: they bend, straighten, and reparameterize the state space through successive non-linear transformations that reduce curvature, contract representational entropy, and reveal the intrinsic geometry of the equilibrium manifold. What appears in standard computational terms as feature extraction or representation learning is, in geometric terms, an iterative search for a coordinate system in which the economic fields $(\Phi, \mathbf{v}, S)$ admit a simple and nearly linear description. The network discovers, rather than assumes, the structure that renders the solution tractable.

Training corresponds to a variational descent on a free-energy functional whose minimizers coincide with the equilibria of the economic model. Stochastic gradient steps, random sampling, and iterative updates become discrete approximations to a continuous relaxation process in which the fields evolve toward the stationary configuration that satisfies all incentive, feasibility, and informational constraints. The monotonic improvement properties observed in practice reflect the geometry of this descent: the flow aligns the scalar potential with intertemporal rewards, the vector field with feasible and optimal adjustment dynamics, and the entropy field with coherent aggregation of uncertainty and beliefs.

This perspective coheres naturally with the broader variational and dynamic framework developed in the preceding sections. Once economic decision rules are understood as field configurations, and once their numerical approximation is interpreted as a curvature-smoothing

process, previously disparate phenomena acquire a unified explanation. Stability of iterative solvers appears as convexity or geodesic regularity of the free-energy landscape. Generalization emerges as the selection of low-entropy, globally coherent field configurations. Canonical structure becomes visible in the evolution of the fields and their conjugate momenta. Multi-scale effects from individual decisions to aggregate distributions find representation in the interplay between local curvature and global entropy.

Seen in this light, the integration of futarchic governance becomes natural rather than extraneous. Predictive markets supply belief-weighted boundary conditions for the fields, while collective value judgments define the welfare functional to be optimized. The futarchic stationarity principle makes this precise: among the family of stationary equilibria permitted by the boundary data, the mechanism selects the policy whose induced field configuration maximizes welfare subject to informational coherence. Policy choice is thus elevated from a comparative statics exercise to a variational selection problem embedded directly within the geometry of the RSVP manifold.

The synthesis accomplished here therefore provides not merely an interpretation of Fernandez-Villaverdes geometric intuition but a rigorous and extensible field-theoretic framework unifying machine learning, equilibrium computation, predictive governance, and high-dimensional economic dynamics. The lamphrodynamic viewpoint clarifies why deep learning succeeds where traditional projection and perturbation methods fail, and it offers new tools for understanding equilibrium selection, stability, bifurcation, robustness, and policy design in complex economic environments. At the same time, it reveals deep structural affinities between economic reasoning, variational physics, predictive markets, and modern representation learning: all seek, in their own languages, the coordinate systems in which the dynamics become simple, the fields become smooth, and the equilibrium becomes manifest.

Appendix: A Field-Theoretic Lagrangian for RSVP Economics with Futarchy Coupling Mechanisms

This appendix presents a formal Lagrangian formulation of dynamic economic models within the Relativistic Scalar Vector Plenum (RSVP) framework. We integrate Robin Hansons principle of *futarchy* —vote on values, bet on beliefs— as a set of gauge-type couplings between value potentials, predictive-belief fields, and policy flows. The result is a unified variational principle governing economic equilibrium, informational aggregation, and decision selection.

A. Field Content

Let $\mathcal{X}$ be the economic state manifold. We define the RSVP fields:

$\Phi(\mathbf{x}, t)$: scalar potential (value field), $\mathbf{v}(\mathbf{x}, t)$: policy vector field, $S(\mathbf{x}, t)$: entropy / belief-distribution field

The field B represents the markets aggregated probabilistic beliefs about future outcomes. Hansons futarchy asserts that B should control *which policy* $\mathbf{v}$ is selected, while Φ encodes the collective welfare target over which society votes.

B. Futarchy Principles as Field Constraints

Hansons motto “vote on values, bet on beliefs” is expressed through:

1. Value Principle:

Φ defines the welfare objective (value votes).

2. Belief Principle:

B encodes predictive accuracy (market bets).

3. Decision Principle:

$\mathbf{v}$ is selected to maximize predicted Φ under B .

Mathematically, this induces a coupling between B and $\mathbf{v}$ analogous to a gauge interaction.

C. Free-Energy and Action Principles

Let the RSVP free-energy functional be:

$$\mathcal{F}[\Phi, \mathbf{v}, S] = \frac{1}{2} \int_{\mathcal{X}} \|T(\Phi, \mathbf{v}, S)\|^2 d\mu,$$

where $T(\Phi, \mathbf{v}, S) = 0$ is the equilibrium operator.

We extend this by adding:

- a **dynamical term** for predictively accurate beliefs,
- a **coupling** between beliefs B and policy flows $\mathbf{v}$,

- a **value alignment potential** coupling Φ to B .

The full *RSVP-Futarchy Action* is:

$$\mathcal{S} = \int dt \int_{\mathcal{X}} \mathcal{L}(\Phi, \mathbf{v}, S, B) d\mu.$$

D. The Lagrangian Density

We propose the following Lagrangian:

$$\boxed{\mathcal{L} = \mathcal{L}_{\text{dyn}} + \mathcal{L}_{\text{equil}} + \mathcal{L}_{\text{belief}} + \mathcal{L}_{\text{futarchy}} + \mathcal{L}_{\text{entropy}}} \quad (8)$$

Each term is defined as follows.

1. Dynamical Term

Lamphrodynamical evolution of fields:

$$\mathcal{L}_{\text{dyn}} = \frac{1}{2} \left(|\partial_t \Phi|^2 + \|\partial_t \mathbf{v}\|^2 + |\partial_t S|^2 + |\partial_t B|^2 \right).$$

2. Equilibrium Constraint Term

Enforces the RSVP economic equilibrium via penalty (free-energy):

$$\mathcal{L}_{\text{equil}} = -\frac{\lambda}{2} \|T(\Phi, \mathbf{v}, S)\|^2.$$

Large λ recovers exact equilibrium.

3. Belief Dynamics Term

Prediction market accuracy emerges from minimizing surprisal:

$$\mathcal{L}_{\text{belief}} = -\frac{\alpha}{2} \|\nabla B\|^2 - \beta B S.$$

Interpretation: - $\|\nabla B\|^2$: smoother belief fields generalize better - BS : beliefs must compete with entropy/noise

4. Futarchy Coupling Term

This encodes actions selected by beliefs to maximize values:

$$\mathcal{L}_{\text{futarchy}} = \gamma B \langle \nabla \Phi, \mathbf{v} \rangle - \frac{\kappa}{2} \|\mathbf{v} - \mathbf{v}_B\|^2.$$

Where: - first term: beliefs B amplify policy-gradient ascent on Φ - second term: $\mathbf{v}_B$ is the belief-implied optimal policy

This is the field-theoretic embodiment of:

$$\text{Best policy} = \text{argmax predicted welfare.}$$

5. Entropy-Field Term

Entropy penalizes sharp, unstable, or overly complex structures:

$$\mathcal{L}_{\text{entropy}} = -\eta |\nabla S|^2 - \xi S \log S.$$

The second term is Shannon entropy.

E. Euler-Lagrange Field Equations

We now vary the action with respect to each field.

Lemma 2 (Euler-Lagrange Equations for RSVP-Futarchy Fields). *The fields satisfy:*

$$\frac{\partial \mathcal{L}}{\partial X} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu X)} \right) = 0, \quad X \in \{\Phi, \mathbf{v}, S, B\}.$$

Solving yields:

Value Field Equation

$$\partial_t^2 \Phi = \lambda \frac{\delta}{\delta \Phi} \|T\|^2 + \gamma \nabla \cdot (B \mathbf{v}).$$

Policy Flow Equation

$$\partial_t^2 \mathbf{v} = \lambda \frac{\delta}{\delta \mathbf{v}} \|T\|^2 + \gamma B \nabla \Phi - \kappa(\mathbf{v} - \mathbf{v}_B).$$

Belief Field Equation

$$\partial_t^2 B = \alpha \Delta B + \beta S + \gamma \langle \nabla \Phi, \mathbf{v} \rangle.$$

Entropy Field Equation

$$\partial_t^2 S = \lambda \frac{\delta}{\delta S} \|T\|^2 + \beta B + 2\eta \Delta S + \xi(1 + \log S).$$

F. Main Result

Theorem 4 (RSVP-Futarchy Equivalence Principle). *A field configuration*

$$(\Phi^*, \mathbf{v}^*, S^*, B^*)$$

is a stationary solution of the futarchy-Lagrangian if and only if:

1. *it satisfies the economic equilibrium condition*

$$T(\Phi^*, \mathbf{v}^*, S^*) = 0,$$

2. *the policy maximizes predicted welfare*

$$\mathbf{v}^* = \arg \max_{\mathbf{v}} \mathbb{E}_{B^*}[\Phi^*],$$

3. *beliefs are predictive-accuracy stationary points,*

4. *and entropy is minimized subject to the above.*

Proof. Stationarity of the action produces the Euler-Lagrange equations above. The λ -term enforces equilibrium. The κ - and γ -terms impose the futarchy decision rule. The remaining terms enforce belief accuracy and entropy minimization. Thus all conditions follow. $\square$

Appendix: Hamiltonian Formulation of the RSVP-Futarchy System

In this appendix we derive the canonical Hamiltonian formulation of the RSVP-Futarchy field theory introduced in the previous appendix. We work with the same field content:

$$\Phi(\mathbf{x}, t), \quad \mathbf{v}(\mathbf{x}, t), \quad S(\mathbf{x}, t), \quad B(\mathbf{x}, t),$$

and the Lagrangian density

$$\mathcal{L} = \mathcal{L}_{\text{dyn}} + \mathcal{L}_{\text{equil}} + \mathcal{L}_{\text{belief}} + \mathcal{L}_{\text{futarchy}} + \mathcal{L}_{\text{entropy}},$$

where

$$\mathcal{L}_{\text{dyn}} = \frac{1}{2} \left(|\partial_t \Phi|^2 + \|\partial_t \mathbf{v}\|^2 + |\partial_t S|^2 + |\partial_t B|^2 \right), \quad (9)$$

$$\mathcal{L}_{\text{equil}} = -\frac{\lambda}{2} \|T(\Phi, \mathbf{v}, S)\|^2, \quad (10)$$

$$\mathcal{L}_{\text{belief}} = -\frac{\alpha}{2} \|\nabla B\|^2 - \beta B S, \quad (11)$$

$$\mathcal{L}_{\text{futarchy}} = \gamma B \langle \nabla \Phi, \mathbf{v} \rangle - \frac{\kappa}{2} \|\mathbf{v} - \mathbf{v}_B\|^2, \quad (12)$$

$$\mathcal{L}_{\text{entropy}} = -\eta |\nabla S|^2 - \xi S \log S. \quad (13)$$

A. Canonical Momenta and Phase Space

We define the conjugate momenta in the standard way:

$$\Pi_X(\mathbf{x}, t) = \frac{\partial \mathcal{L}}{\partial(\partial_t X(\mathbf{x}, t))}, \quad X \in \{\Phi, \mathbf{v}, S, B\}.$$

Since only $\mathcal{L}_{\text{dyn}}$ depends on time derivatives, the canonical momenta are:

$$\Pi_\Phi = \frac{\partial \mathcal{L}}{\partial(\partial_t \Phi)} = \partial_t \Phi, \quad (14)$$

$$\mathbf{\Pi}_\mathbf{v} = \frac{\partial \mathcal{L}}{\partial(\partial_t \mathbf{v})} = \partial_t \mathbf{v}, \quad (15)$$

$$\Pi_S = \frac{\partial \mathcal{L}}{\partial(\partial_t S)} = \partial_t S, \quad (16)$$

$$\Pi_B = \frac{\partial \mathcal{L}}{\partial(\partial_t B)} = \partial_t B. \quad (17)$$

Definition 4 (RSVP–Futarchy Phase Space). *The canonical phase space consists of the fields and their conjugate momenta:*

$$(\Phi, \Pi_\Phi; \mathbf{v}, \mathbf{\Pi}_\mathbf{v}; S, \Pi_S; B, \Pi_B).$$

In this formulation, the lamphrodynamic evolution is a Hamiltonian flow on this infinite-dimensional phase space.

B. Hamiltonian Density

The Hamiltonian density is defined as

$$\mathcal{H} = \Pi_\Phi \partial_t \Phi + \mathbf{\Pi}_\mathbf{v} \cdot \partial_t \mathbf{v} + \Pi_S \partial_t S + \Pi_B \partial_t B - \mathcal{L}.$$

Substituting $\partial_t \Phi = \Pi_\Phi$, etc., we obtain

$$\Pi_\Phi \partial_t \Phi + \mathbf{\Pi}_\mathbf{v} \cdot \partial_t \mathbf{v} + \Pi_S \partial_t S + \Pi_B \partial_t B = |\Pi_\Phi|^2 + \|\mathbf{\Pi}_\mathbf{v}\|^2 + |\Pi_S|^2 + |\Pi_B|^2 = 2 \mathcal{L}_{\text{dyn}}. \quad (18)$$

Therefore,

$$\mathcal{H} = 2\mathcal{L}_{\text{dyn}} - (\mathcal{L}_{\text{dyn}} + \mathcal{L}_{\text{equil}} + \mathcal{L}_{\text{belief}} + \mathcal{L}_{\text{futarchy}} + \mathcal{L}_{\text{entropy}}) \quad (19)$$

$$= \mathcal{L}_{\text{dyn}} - \mathcal{L}_{\text{equil}} - \mathcal{L}_{\text{belief}} - \mathcal{L}_{\text{futarchy}} - \mathcal{L}_{\text{entropy}}. \quad (20)$$

Using the explicit forms of each term:

$$\mathcal{H} = \frac{1}{2} \left(|\Pi_\Phi|^2 + \|\mathbf{\Pi}_\mathbf{v}\|^2 + |\Pi_S|^2 + |\Pi_B|^2 \right) + \frac{\lambda}{2} \|T(\Phi, \mathbf{v}, S)\|^2 \quad (21)$$

$$+ \frac{\alpha}{2} \|\nabla B\|^2 + \beta BS \quad (22)$$

$$- \gamma B \langle \nabla \Phi, \mathbf{v} \rangle + \frac{\kappa}{2} \|\mathbf{v} - \mathbf{v}_B\|^2 \quad (23)$$

$$+ \eta |\nabla S|^2 + \xi S \log S. \quad (24)$$

Definition 5 (RSVP–Futarchy Hamiltonian). *The Hamiltonian functional is*

$$H[\Phi, \mathbf{v}, S, B; \Pi_\Phi, \mathbf{\Pi}_\mathbf{v}, \Pi_S, \Pi_B] = \int_{\mathcal{X}} \mathcal{H} d\mu.$$

This H plays the role of a generalized free energy, combining: kinetic contributions from the momenta, equilibrium violations, belief curvature, futarchy couplings, and entropy penalties.

C. Hamiltons Equations

The canonical Hamilton equations take the form

$$\partial_t X = \frac{\delta H}{\delta \Pi_X}, \quad (25)$$

$$\partial_t \Pi_X = -\frac{\delta H}{\delta X}, \quad (26)$$

for each field $X \in \{\Phi, \mathbf{v}, S, B\}$.

1. Value Field

$$\partial_t \Phi = \frac{\delta H}{\delta \Pi_\Phi} = \Pi_\Phi, \quad (27)$$

$$\partial_t \Pi_\Phi = -\frac{\delta H}{\delta \Phi}. \quad (28)$$

The second equation encodes the effects of:

- equilibrium curvature via $\lambda \|T\|^2$,
- futarchy coupling via $-\gamma B \langle \nabla(\cdot), \mathbf{v} \rangle$.

2. Policy Field

$$\partial_t \mathbf{v} = \frac{\delta H}{\delta \Pi_{\mathbf{v}}} = \Pi_{\mathbf{v}}, \quad (29)$$

$$\partial_t \Pi_{\mathbf{v}} = -\frac{\delta H}{\delta \mathbf{v}}. \quad (30)$$

The functional derivative includes:

- equilibrium penalties,
- a futarchy term $+\gamma B \nabla \Phi$,
- a restoring term $-\kappa(\mathbf{v} - \mathbf{v}_B)$.

3. Belief Field

$$\partial_t B = \frac{\delta H}{\delta \Pi_B} = \Pi_B, \quad (31)$$

$$\partial_t \Pi_B = -\frac{\delta H}{\delta B}. \quad (32)$$

The potential term in H gives:

$$\frac{\delta H}{\delta B} = \beta S - \gamma \langle \nabla \Phi, \mathbf{v} \rangle - \alpha \Delta B,$$

so

$$\partial_t^2 B = \alpha \Delta B + \beta S + \gamma \langle \nabla \Phi, \mathbf{v} \rangle,$$

matching the Euler–Lagrange equation.

4. Entropy Field

$$\partial_t S = \frac{\delta H}{\delta \Pi_S} = \Pi_S, \quad (33)$$

$$\partial_t \Pi_S = -\frac{\delta H}{\delta S}. \quad (34)$$

The potential includes:

$$\frac{\delta H}{\delta S} = \lambda \frac{\delta}{\delta S} \|T\|^2 + \beta B + 2\eta \Delta S + \xi(1 + \log S),$$

giving the same entropy evolution as in the Lagrangian formulation.

Proposition 2 (Equivalence with Euler–Lagrange Dynamics). *The Hamiltonian equations derived above are equivalent to the Euler–Lagrange field equations obtained from the RSVP–Futarchy Lagrangian.*

Proof. Direct verification: for each field, the Hamiltonian evolution reproduces the second-order time equations obtained from the Euler–Lagrange formalism, given the identification of momenta $\Pi_X = \partial_t X$. $\square$

D. Conceptual Interpretation

The Hamiltonian formulation emphasizes that the RSVP–Futarchy system is a *phase-space dynamical system* in which:

- the fields $(\Phi, \mathbf{v}, S, B)$ encode welfare, policy, entropy, and collective beliefs;
- the conjugate momenta represent the “lamphrodynamic velocities” of these structures;
- the Hamiltonian H acts as a generalized free energy, combining equilibrium constraints, prediction markets, and entropy penalties;
- futarchy couplings appear as cross-terms in the potential energy that direct policy flows along those trajectories that markets predict will maximize welfare.

Stationary points of H correspond to fully self-consistent economic equilibria in which values, beliefs, policies, and entropy are jointly aligned. The Hamiltonian view thus complements the Lagrangian field-theoretic picture and provides a natural foundation for canonical quantization or stochastic extensions of predictive market dynamics.

Appendix: Poisson Brackets and Symplectic Structure for RSVP–Futarchy Fields

In this appendix we endow the RSVP–Futarchy Hamiltonian system with a canonical symplectic structure and the associated Poisson bracket formalism. This makes explicit that the lamphrodynamic evolution of welfare, policy, entropy, and belief fields is a Hamiltonian flow on an infinite-dimensional phase space.

We continue with the field content and canonical momenta introduced in the Hamiltonian appendix:

$$\Phi(\mathbf{x}, t), \quad \mathbf{v}(\mathbf{x}, t), \quad S(\mathbf{x}, t), \quad B(\mathbf{x}, t),$$

with conjugate momenta

$$\Pi_\Phi(\mathbf{x}, t), \quad \Pi_{\mathbf{v}}(\mathbf{x}, t), \quad \Pi_S(\mathbf{x}, t), \quad \Pi_B(\mathbf{x}, t),$$

and Hamiltonian functional

$$H[\Phi, \mathbf{v}, S, B; \Pi_\Phi, \Pi_{\mathbf{v}}, \Pi_S, \Pi_B] = \int_{\mathcal{X}} \mathcal{H} d\mu.$$

A. Canonical Poisson Brackets

We treat the fields and their conjugate momenta as coordinates on an infinite-dimensional phase space and define the canonical Poisson brackets in the standard functional sense.

Let F and G be functionals of the fields and momenta. The *canonical Poisson bracket* is defined as

$$\{F, G\} = \int_{\mathcal{X}} \left[\frac{\delta F}{\delta \Phi(\mathbf{x})} \frac{\delta G}{\delta \Pi_\Phi(\mathbf{x})} - \frac{\delta F}{\delta \Pi_\Phi(\mathbf{x})} \frac{\delta G}{\delta \Phi(\mathbf{x})} \right] d\mu(\mathbf{x}) + \cdots, \quad (35)$$

with analogous contributions from the other field–momentum pairs:

$$(\mathbf{v}, \Pi_{\mathbf{v}}), \quad (S, \Pi_S), \quad (B, \Pi_B).$$

More explicitly,

$$\{F, G\} = \int_{\mathcal{X}} \left[\frac{\delta F}{\delta \Phi} \frac{\delta G}{\delta \Pi_{\Phi}} - \frac{\delta F}{\delta \Pi_{\Phi}} \frac{\delta G}{\delta \Phi} \right] d\mu \quad (36)$$

$$+ \int_{\mathcal{X}} \left[\frac{\delta F}{\delta v^i} \frac{\delta G}{\delta \Pi_{v^i}} - \frac{\delta F}{\delta \Pi_{v^i}} \frac{\delta G}{\delta v^i} \right] d\mu \quad (37)$$

$$+ \int_{\mathcal{X}} \left[\frac{\delta F}{\delta S} \frac{\delta G}{\delta \Pi_S} - \frac{\delta F}{\delta \Pi_S} \frac{\delta G}{\delta S} \right] d\mu \quad (38)$$

$$+ \int_{\mathcal{X}} \left[\frac{\delta F}{\delta B} \frac{\delta G}{\delta \Pi_B} - \frac{\delta F}{\delta \Pi_B} \frac{\delta G}{\delta B} \right] d\mu, \quad (39)$$

where repeated spatial indices i are summed and we suppress explicit $\mathbf{x}$ -dependence for readability.

Definition 6 (Canonical Equal-Time Poisson Brackets). *The fundamental brackets between the fields and their conjugate momenta at equal time are:*

$$\{\Phi(\mathbf{x}), \Pi_{\Phi}(\mathbf{y})\} = \delta(\mathbf{x} - \mathbf{y}), \quad \{\Phi(\mathbf{x}), \Phi(\mathbf{y})\} = 0, \quad \{\Pi_{\Phi}(\mathbf{x}), \Pi_{\Phi}(\mathbf{y})\} = 0, \quad (40)$$

$$\{v^i(\mathbf{x}), \Pi_{v^i}(\mathbf{y})\} = \delta^i_j \delta(\mathbf{x} - \mathbf{y}), \quad \{v^i(\mathbf{x}), v^j(\mathbf{y})\} = 0, \quad \{\Pi_{v^i}(\mathbf{x}), \Pi_{v^j}(\mathbf{y})\} = 0, \quad (41)$$

$$\{S(\mathbf{x}), \Pi_S(\mathbf{y})\} = \delta(\mathbf{x} - \mathbf{y}), \quad \{S(\mathbf{x}), S(\mathbf{y})\} = 0, \quad \{\Pi_S(\mathbf{x}), \Pi_S(\mathbf{y})\} = 0, \quad (42)$$

$$\{B(\mathbf{x}), \Pi_B(\mathbf{y})\} = \delta(\mathbf{x} - \mathbf{y}), \quad \{B(\mathbf{x}), B(\mathbf{y})\} = 0, \quad \{\Pi_B(\mathbf{x}), \Pi_B(\mathbf{y})\} = 0, \quad (43)$$

with all cross-brackets between distinct fields vanishing.

These Poisson brackets satisfy the usual antisymmetry, bilinearity, Leibniz rule, and Jacobi identity, and thus equip the function space with a Lie-algebra structure.

B. Symplectic 2-Form

The canonical brackets arise from a symplectic 2-form on phase space.

Definition 7 (RSVP–Futarchy Symplectic Form). *The symplectic form on the phase space of fields is*

$$\Omega = \int_{\mathcal{X}} \left(\delta \Pi_{\Phi} \wedge \delta \Phi + \delta \Pi_{\mathbf{v}} \wedge \delta \mathbf{v} + \delta \Pi_S \wedge \delta S + \delta \Pi_B \wedge \delta B \right) d\mu. \quad (44)$$

Here δ denotes the exterior derivative on the infinite-dimensional field space, and $\wedge$ is the wedge product. Non-degeneracy and closedness ($d\Omega = 0$) ensure that (phase space, Ω) is a symplectic manifold in the functional sense.

Lemma 3 (Poisson Brackets from Ω). *The Poisson bracket of functionals F and G can be written as*

$$\{F, G\} = \Omega(X_F, X_G),$$

where X_F is the Hamiltonian vector field associated with F , defined implicitly by

$$\iota_{X_F}\Omega = \delta F.$$

Proof. Standard symplectic geometry argument: the definition of X_F via $\iota_{X_F}\Omega = \delta F$ implies that for any G ,

$$\{F, G\} := \Omega(X_F, X_G) = \delta F(X_G),$$

which reproduces the functional expression given earlier. $\square$

Thus the canonical brackets and symplectic form are equivalent ways of encoding the same geometric structure.

C. Hamiltonian Flow as Lamphrodynamic Evolution

With the symplectic form and Poisson bracket defined, the evolution of any functional F of the fields and momenta under the Hamiltonian

$$H = \int_{\mathcal{X}} \mathcal{H} d\mu$$

is given by

$$\frac{dF}{dt} = \{F, H\}.$$

In particular, for the fields themselves:

$$\partial_t \Phi = \{\Phi, H\} = \frac{\delta H}{\delta \Pi_\Phi}, \quad \partial_t \Pi_\Phi = \{\Pi_\Phi, H\} = -\frac{\delta H}{\delta \Phi}, \quad (45)$$

$$\partial_t \mathbf{v} = \{\mathbf{v}, H\} = \frac{\delta H}{\delta \Pi_{\mathbf{v}}}, \quad \partial_t \Pi_{\mathbf{v}} = \{\Pi_{\mathbf{v}}, H\} = -\frac{\delta H}{\delta \mathbf{v}}, \quad (46)$$

$$\partial_t S = \{S, H\} = \frac{\delta H}{\delta \Pi_S}, \quad \partial_t \Pi_S = \{\Pi_S, H\} = -\frac{\delta H}{\delta S}, \quad (47)$$

$$\partial_t B = \{B, H\} = \frac{\delta H}{\delta \Pi_B}, \quad \partial_t \Pi_B = \{\Pi_B, H\} = -\frac{\delta H}{\delta B}. \quad (48)$$

Proposition 3 (Lamphrodynamic Flow as Hamiltonian Flow). *The lamphrodynamic evolution derived from the Lagrangian formulation is identical to the Hamiltonian flow generated by H with respect to the canonical Poisson brackets.*

Proof. Substituting the Hamiltonian density $\mathcal{H}$ into the functional derivatives above reproduces the Euler–Lagrange equations for each field, given the identification $\Pi_X = \partial_t X$. This is the standard equivalence between the Lagrangian and Hamiltonian pictures. $\square$

Thus the RSVP–Futarchy system is a fully symplectic dynamical system, with lamphrodynamic evolution interpreted as a Hamiltonian flow on the space of welfare, policy, entropy, and belief fields.

D. Remarks on Constraints and Futarchy Couplings

In the present construction we have taken the field variables and momenta as unconstrained canonical coordinates. In more refined models:

- the equilibrium condition $T(\Phi, \mathbf{v}, S) = 0$ can be implemented as a *constraint surface* in phase space;
- the futarchy relationship $\mathbf{v} \approx \mathbf{v}_B(\Phi, B)$ may introduce additional (possibly second-class) constraints;
- value-gauge symmetries (redefining Φ by additive constants or linear transformations) may appear as first-class constraints generating gauge orbits.

A full Dirac constraint analysis would then identify reduced phase spaces and modified (Dirac) brackets. For the purposes of this essay, however, the canonical Poisson structure suffices to demonstrate that the RSVP–Futarchy fields form a legitimate Hamiltonian field theory, ready for possible quantization or stochastic deformation.

Appendix: Path and Functional Integrals for the RSVP–Futarchy Field Theory

This appendix provides a functional-integral formulation of the RSVPFutarchy system. In this picture, welfare fields, policy fields, entropy fields, and belief fields are integrated over all possible configurations weighted by the futarchy Lagrangian. The result is a field-theoretic interpretation of prediction markets as information- aggregating functional integrators.

A. The Action and Functional Measure

We begin with the Lagrangian density

$$\mathcal{L} = \mathcal{L}_{\text{dyn}} + \mathcal{L}_{\text{equil}} + \mathcal{L}_{\text{belief}} + \mathcal{L}_{\text{futarchy}} + \mathcal{L}_{\text{entropy}},$$

with explicit forms given in the earlier appendix.

The action is

$$S[\Phi, \mathbf{v}, S, B] = \int dt \int_{\mathcal{X}} \mathcal{L}(\Phi, \mathbf{v}, S, B) d\mu.$$

The formal functional measure is

$$\mathcal{D}\Phi \mathcal{D}\mathbf{v} \mathcal{D}S \mathcal{D}B,$$

which integrates over all histories of the fields.

B. The RSVP–Futarchy Generating Functional

The central object is the generating functional:

$$Z[J_\Phi, J_\mathbf{v}, J_S, J_B] = \int \mathcal{D}\Phi \mathcal{D}\mathbf{v} \mathcal{D}S \mathcal{D}B \exp \left[-S[\Phi, \mathbf{v}, S, B] + \int J \cdot X \right]. \quad (49)$$

Here:

$$J \cdot X = \int dt \int_{\mathcal{X}} (J_\Phi \Phi + J_\mathbf{v} \cdot \mathbf{v} + J_S S + J_B B) d\mu.$$

We adopt the Euclidean convention ($\exp[-S]$) for stability and interpretability.

Correlation functions follow by functional differentiation:

$$\langle X_1(\mathbf{x}_1, t_1) \cdots X_n(\mathbf{x}_n, t_n) \rangle = \frac{\delta^n Z}{\delta J_1 \cdots \delta J_n} \Big|_{J=0}.$$

C. Integrating Out the Belief Field B

A central feature of futarchy is that predictive markets aggregate information into a single, coherent belief distribution. This property emerges naturally from the functional integral.

We isolate all terms involving B in the Euclidean action:

$$S_B = \int dt \int_{\mathcal{X}} \left[\frac{1}{2} |\partial_t B|^2 + \frac{\alpha}{2} \|\nabla B\|^2 + \beta B S - \gamma B \langle \nabla \Phi, \mathbf{v} \rangle \right] d\mu. \quad (50)$$

This is a quadratic functional of B with linear source:

$$J_B^{\text{eff}} = \beta S - \gamma \langle \nabla \Phi, \mathbf{v} \rangle.$$

Thus the belief integral is Gaussian:

$$\int \mathcal{D}B \exp[-S_B] = (\det \mathcal{O}_B)^{-1/2} \exp \left[\frac{1}{2} \int J_B^{\text{eff}} \mathcal{O}_B^{-1} J_B^{\text{eff}} \right], \quad (51)$$

where

$$\mathcal{O}_B = -\partial_t^2 + \alpha \Delta$$

is the belief-field kinetic operator.

Theorem 5 (Futarchy as Optimal Information Aggregation). *Integrating out the belief field produces a nonlocal effective action that encodes optimal prediction aggregation:*

$$S_{\text{eff}} = S[\Phi, \mathbf{v}, S] - \frac{1}{2} \int J_B^{\text{eff}} \mathcal{O}_B^{-1} J_B^{\text{eff}}.$$

Proof. Gaussian functional integration. The inverse operator $\mathcal{O}_B^{-1}$ is the Greens function that aggregates information from all points in spacetime. $\square$

This term is explicitly nonlocal:

$$\mathcal{O}_B^{-1}(\mathbf{x}, t; \mathbf{y}, t') \sim \text{propagator connecting all prediction sources.}$$

Economically, it captures how speculative markets pool information, compare predictions globally, and feed back into policy flows.

D. Effective Action After Integrating Out B

The effective action is:

$$S_{\text{eff}} = \int dt \int_{\mathcal{X}} [\mathcal{L}_{\text{dyn}} + \mathcal{L}_{\text{equil}} + \mathcal{L}_{\text{entropy}}] d\mu + S_{\text{belief-induced}}[\Phi, \mathbf{v}, S],$$

with

$$S_{\text{belief-induced}} = -\frac{1}{2} \int dt dt' \int_{\mathcal{X}} d\mu \int_{\mathcal{X}} d\mu' J_B^{\text{eff}}(x) \mathcal{O}_B^{-1}(x, x') J_B^{\text{eff}}(x').$$

This contains cross-terms of the form:

$$\frac{1}{2} \gamma^2 \langle \nabla \Phi, \mathbf{v} \rangle(x) \mathcal{O}_B^{-1}(x, x') \langle \nabla \Phi, \mathbf{v} \rangle(x'), \quad (52)$$

$$- \gamma \beta \langle \nabla \Phi, \mathbf{v} \rangle(x) \mathcal{O}_B^{-1}(x, x') S(x'), \quad (53)$$

$$\frac{1}{2} \beta^2 S(x) \mathcal{O}_B^{-1}(x, x') S(x'). \quad (54)$$

Definition 8 (Futarchy Kernel). *Define the kernel*

$$K(x, x') := \mathcal{O}_B^{-1}(x, x')$$

as the belief-propagation operator.

Then

$$S_{\text{belief-induced}} = -\frac{1}{2} \iint J_B^{\text{eff}}(x) K(x, x') J_B^{\text{eff}}(x') dx dx'.$$

This confirms that the market-belief field creates long-range interactions across the economic manifold, exactly as futarchy predicts.

E. Probability Amplitude for Policy Paths

The probability amplitude for a policy trajectory $\mathbf{v}(t)$ is:

$$\mathcal{P}[\mathbf{v}] \propto \int \mathcal{D}\Phi \mathcal{D}S \exp[-S_{\text{eff}}(\Phi, \mathbf{v}, S)].$$

The most probable trajectories minimize the effective action:

$$\frac{\delta S_{\text{eff}}}{\delta \mathbf{v}} = 0,$$

which yields the futarchy-adjusted EulerLagrange equations.

This is the field-theoretic proof that:

• *Prediction markets select the policies that maximize expected value.*

F. Functional Integral Interpretation of Futarchy

This framework yields the following interpretation:

- Welfare potential Φ is what society votes on.
- Policy field $\mathbf{v}$ is what society chooses (or what the system evolves toward).
- The belief field B is integrated out, leaving behind a nonlocal kernel $K = \mathcal{O}^{-1}$ that aggregates all predictive information.
- Policy is chosen to minimize the effective action, i.e., to maximize expected welfare under market-generated probabilistic beliefs.

Thus futarchy naturally emerges as:

Value selection via variational optimization and belief aggregation via functional integration.

This gives a clean mathematical foundation for the slogan:

Vote on values; bet on beliefs.

Appendix: Renormalization of the Belief Field and Scale-Dependent Futarchy Couplings

In this appendix we analyze the RSVP–Futarchy field theory from a Wilsonian renormalization-group (RG) viewpoint. The central idea is that the belief field B acts as a scale-dependent

information aggregator: short-scale fluctuations represent noisy, local bets, while long-scale modes encode robust, macro-level predictions. By integrating out high-frequency components of B , we obtain an effective theory with scale-dependent couplings, providing a renormalization interpretation of futarchy.

A. Momentum-Shell Decomposition of the Belief Field

For simplicity, we work in a flat spatial manifold with coordinates $\mathbf{x} \in \mathbb{R}^d$ and use a Fourier representation. The extension to curved $\mathcal{X}$ uses eigenmodes of the Laplacian.

Let

$$B(\mathbf{x}, t) = \int \frac{d^d k}{(2\pi)^d} e^{i\mathbf{k} \cdot \mathbf{x}} \tilde{B}(\mathbf{k}, t).$$

We introduce a UV cutoff Λ and decompose modes into:

$$\tilde{B}_{<}(\mathbf{k}, t) : \|\mathbf{k}\| \leq \Lambda/b, \quad (55)$$

$$\tilde{B}_{>}(\mathbf{k}, t) : \Lambda/b < \|\mathbf{k}\| \leq \Lambda, \quad (56)$$

for some rescaling factor $b > 1$. Correspondingly,

$$B = B_{<} + B_{>}.$$

Definition 9 (Belief Field Mode Split). *The low-frequency component $B_{<}$ encodes large-scale, aggregated predictive structure, while $B_{>}$ encodes short-scale, local fluctuations in beliefs.*

The Wilsonian RG step integrates out $B_{>}$ to obtain an effective action $S_{\text{eff}}[\Phi, \mathbf{v}, S, B_{<}; \ell]$ at scale $\ell = \log b$.

B. Belief Sector of the Action

Recall the B -dependent part of the Euclidean action:

$$S_B[B; \Phi, \mathbf{v}, S] = \int dt \int_{\mathcal{X}} \left[\frac{1}{2} |\partial_t B|^2 + \frac{\alpha}{2} \|\nabla B\|^2 + \beta B S - \gamma B \langle \nabla \Phi, \mathbf{v} \rangle \right] d\mu.$$

In Fourier space (suppressing time and indices),

$$S_B = \frac{1}{2} \int_{\|\mathbf{k}\| \leq \Lambda} \tilde{B}(-\mathbf{k}) \mathcal{O}_B(\mathbf{k}) \tilde{B}(\mathbf{k}) d^d k + \int_{\|\mathbf{k}\| \leq \Lambda} \tilde{B}(-\mathbf{k}) \tilde{J}_B(\mathbf{k}) d^d k,$$

with

$$\mathcal{O}_B(\mathbf{k}) = \omega^2 + \alpha \|\mathbf{k}\|^2, \quad \tilde{J}_B(\mathbf{k}) = \widetilde{\beta S} - \gamma \widetilde{\langle \nabla \Phi, \mathbf{v} \rangle}.$$

We now perform a Wilsonian integration of $B_>$.

C. Wilsonian Integration of High-Frequency Belief Modes

Decompose

$$\tilde{B}(\mathbf{k}) = \tilde{B}_<(\mathbf{k}) \theta(\Lambda/b - \|\mathbf{k}\|) + \tilde{B}_>(\mathbf{k}) \theta(\|\mathbf{k}\| - \Lambda/b) \theta(\Lambda - \|\mathbf{k}\|).$$

The partition function becomes

$$Z = \int \mathcal{D}\Phi \mathcal{D}\mathbf{v} \mathcal{D}S \int \mathcal{D}B_< \mathcal{D}B_> e^{-S[\Phi, \mathbf{v}, S, B_< + B_>}.$$

The Wilsonian step defines an effective action at scale Λ/b via

$$e^{-S_{\text{eff}}[\Phi, \mathbf{v}, S, B_<; \Lambda/b]} = \int \mathcal{D}B_> e^{-S_B[B_< + B_>; \Phi, \mathbf{v}, S]} e^{-S_{\text{rest}}[\Phi, \mathbf{v}, S]},$$

where S_{rest} contains all other fields contributions.

Because S_B is quadratic in B , the integral over $B_>$ is again Gaussian, yielding a correction term to the effective action for $B_<$:

$$S_{\text{eff}}[\Phi, \mathbf{v}, S, B_<] = S_B[B_<; \Phi, \mathbf{v}, S] + \Delta S_{\Lambda \rightarrow \Lambda/b}[\Phi, \mathbf{v}, S] + S_{\text{rest}}[\Phi, \mathbf{v}, S]. \quad (57)$$

The correction term ΔS arises from the contraction of the source $\tilde{J}_B$ with the propagator over the high-momentum shell:

$$\Delta S_{\Lambda \rightarrow \Lambda/b} = -\frac{1}{2} \int_{\Lambda/b < \|\mathbf{k}\| \leq \Lambda} \frac{|\tilde{J}_B(\mathbf{k})|^2}{\mathcal{O}_B(\mathbf{k})} d^d k. \quad (58)$$

This correction renormalizes the couplings that multiply S and $\langle \nabla \Phi, \mathbf{v} \rangle$ in the low-energy theory.

D. Running Couplings and Beta Functions

At leading order, we can treat $\tilde{J}_B$ as small and expand ΔS to extract the induced running of parameters. The belief sector couplings (α, β, γ) now become scale-dependent:

$$\alpha \rightarrow \alpha(\ell), \quad \beta \rightarrow \beta(\ell), \quad \gamma \rightarrow \gamma(\ell),$$

where $\ell = \log(\Lambda/\Lambda_{\text{eff}})$.

Definition 10 (RG Flow of Futarchy Couplings). *We define beta functions by*

$$\frac{d\alpha}{d\ell} = \beta_\alpha(\alpha, \beta, \gamma, \dots), \quad \frac{d\beta}{d\ell} = \beta_\beta(\alpha, \beta, \gamma, \dots), \quad \frac{d\gamma}{d\ell} = \beta_\gamma(\alpha, \beta, \gamma, \dots).$$

Qualitatively:

- β_α captures how the “belief stiffness” (sensitivity to spatial variation) changes with scale.
- β_β measures how strongly beliefs couple to entropy/noise at different resolutions.
- β_γ measures how strongly beliefs amplify the gradient-aligned policy flow $\langle \nabla \Phi, \mathbf{v} \rangle$.

At one-loop (Gaussian) level, the corrections come from integrals of the form

$$\int_{\Lambda/b < \|\mathbf{k}\| \leq \Lambda} \frac{d^d k}{(2\pi)^d} \frac{1}{\omega^2 + \alpha \|\mathbf{k}\|^2} \sim C_d \Lambda^{d-2}$$

for some dimension-dependent constant C_d , leading to power-law or logarithmic running depending on d .

Proposition 4 (Scale-Dependent Predictive Market Regimes). *If the beta functions drive $\alpha(\ell)$, $\beta(\ell)$, or $\gamma(\ell)$ toward fixed points as $\ell \rightarrow \infty$ (IR limit), the futarchy dynamics exhibit scale-invariant prediction and policy selection. Otherwise, crossovers between distinct predictive regimes occur as one moves from micro to macro scales.*

Proof. Standard RG reasoning: fixed points correspond to scale invariance of the effective action; absence of fixed points implies flow between regimes with different effective couplings. $\square$

E. Interpretation: Information Coarse-Graining in Futarchy

The RG picture provides a natural interpretation of belief aggregation:

- Integrating out $B_{>}$ corresponds to ignoring short-wavelength, local, or idiosyncratic speculative noise and retaining only the coarse-grained, robust belief structure.
- The scale dependence of (α, β, γ) describes how belief stiffness, noise sensitivity, and policy leverage change as we move from micro-markets to macro-markets.
- RG fixed points correspond to *stable predictive regimes*: scales at which the aggregate futarchy dynamics look self-similar.

This yields a renormalization-based restatement of Hansons core intuition:

Prediction markets that operate across scales increasingly filter out local noise and converge to stable, scale-invariant aggregated beliefs, which in turn guide macro policy.

F. Toward an RG Classification of Economic Prediction Regimes

In principle, one could classify economic environments by the structure of their futarchy RG flows:

- **Efficient-futarchy regime:** $\gamma(\ell) \rightarrow \gamma^* > 0$ and $\beta(\ell) \rightarrow 0$ as $\ell \rightarrow \infty$, implying highly informative beliefs and low noise at large scales.
- **Noisy-futarchy regime:** $\beta(\ell)$ grows while $\gamma(\ell)$ flows to zero, implying that noise dominates and predictive coupling to policy vanishes at macro scales.
- **Critical-futarchy regime:** multiple couplings flow to a nontrivial fixed point, suggesting scale-free fluctuations in beliefs and policy, akin to critical phenomena.

A full analysis would derive explicit beta functions for specific economic microfoundations and belief-update mechanisms. This appendix establishes the conceptual backbone for such a program: futarchy, in the RSVP field-theoretic sense, is an RG process that progressively coarse-grains speculative information while reshaping the couplings between belief, value, policy, and entropy fields.

Appendix: RSVP–Futarchy Formulation of an Aiyagari Economy

This appendix specializes the RSVP–Futarchy framework to a canonical incomplete-markets (Aiyagari) model. We explicitly define the economic state space, map it to RSVP fields $(\Phi, \mathbf{v}, S, B)$, construct the equilibrium operator $T(\Phi, \mathbf{v}, S)$, and then insert this operator into the RSVP–Futarchy Lagrangian.

A. Baseline Aiyagari Model

Consider a continuum of ex-ante identical households indexed by i , each solving:

$$\max_{\{c_{it}, a_{i,t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_{it}),$$

subject to

$$c_{it} + a_{i,t+1} = (1 + r_t)a_{it} + w_t z_{it}, \quad (59)$$

$$a_{i,t+1} \geq \underline{a}, \quad (60)$$

where:

- a_{it} : individual assets,
- $z_{it} \in \mathcal{Z}$: idiosyncratic productivity shock following a Markov chain,
- r_t : interest rate,
- w_t : wage rate,
- $u(\cdot)$: period utility (e.g. CRRA).

The aggregate capital stock K_t is determined by the stationary distribution of assets across households:

$$K_t = \int a d\mu_t(a, z),$$

where μ_t is the cross-sectional distribution over (a, z) .

Firms are competitive with production function $F(K_t, L_t)$, with L_t denoting effective labor. Prices satisfy:

$$r_t = F_K(K_t, L_t) - \delta, \quad w_t = F_L(K_t, L_t).$$

A stationary recursive competitive equilibrium consists of decision rules, prices, and a stationary wealth distribution satisfying household optimality, firm optimality, and market clearing.

B. State Space and RSVP Fields

We define the individual state as

$$x = (a, z, K),$$

where K is the aggregate capital stock (or belief about it). The economic state manifold is:

$$\mathcal{X} = [\underline{a}, \bar{a}] \times \mathcal{Z} \times [0, \bar{K}].$$

We embed this into RSVP fields:

- **Value Field** $\Phi(x)$: the households lifetime value function given (a, z, K) :

$$\Phi(a, z, K) \approx V(a, z, K).$$

- **Policy Field** $\mathbf{v}(x)$: a vector giving the optimal controls. For a simple Aiyagari model,

$$\mathbf{v}(a, z, K) = \begin{pmatrix} c^*(a, z, K) \\ a'^*(a, z, K) \end{pmatrix},$$

representing consumption and next-period assets.

- **Entropy Field** $S(x)$: an entropy density over (a, z, K) , encoding the complexity or dispersion of the cross-sectional distribution and beliefs.
- **Belief Field** $B(x)$: a futarchy prediction field representing market-implied beliefs about future aggregates (e.g. K', r', w') contingent on policies.

Thus the RSVP fields live on $\mathcal{X}$ and encode both micro and macro structure.

C. Equilibrium Operator for the Aiyagari Model

We now construct the operator $T(\Phi, \mathbf{v}, S)$ whose zeroes characterize equilibrium.

1. Bellman Residual

Let $\mathcal{T}$ be the Bellman operator:

$$(\mathcal{T}\Phi)(a, z, K) = \max_{a'} \left\{ u((1 + r(K))a + w(K)z - a') + \beta \mathbb{E}[\Phi(a', z', K') \mid z] \right\}.$$

In a stationary equilibrium with constant K , we have $K' = K$, but in general we allow aggregate dynamics $K' = G(\mu)$ or $K' = G(\Phi, \mathbf{v}, S)$.

The value function residual is:

$$R_V(a, z, K) = \Phi(a, z, K) - (\mathcal{T}\Phi)(a, z, K).$$

2. Policy Consistency Residual

The policy field $\mathbf{v}$ must attain the Bellman maximizer. Let:

$$a'_\Phi(a, z, K) = \arg \max_{a'} \left\{ u((1 + r(K))a + w(K)z - a') + \beta \mathbb{E}[\Phi(a', z', K') \mid z] \right\}.$$

Define the policy residual as:

$$R_{\mathbf{v}}(a, z, K) = \begin{pmatrix} c(a, z, K) - ((1 + r(K))a + w(K)z - a'(a, z, K)) \\ a'(a, z, K) - a'_\Phi(a, z, K) \end{pmatrix},$$

where

$$\mathbf{v}(a, z, K) = \begin{pmatrix} c(a, z, K) \\ a'(a, z, K) \end{pmatrix}.$$

3. Distribution and Market-Clearing Residual

Let $\mu(a, z, K)$ denote a density-compatible representation of the state distribution. The Kolmogorov (law-of-motion) residual R_μ captures the mismatch between the implied stationary distribution under policy $\mathbf{v}$ and the actual distribution μ . At a high level:

$$R_\mu(a, z, K) = \mu(a, z, K) - \mathcal{P}_\mathbf{v}[\mu](a, z, K),$$

where $\mathcal{P}_\mathbf{v}$ is the operator that pushes μ forward via the transition $(a, z) \rightarrow (a'(a, z, K), z')$.

The aggregate capital residual is:

$$R_K(K, \mu) = K - \int a d\mu(a, z, K).$$

4. Entropy Residual

We define an entropy residual capturing the discrepancy between the actual distribution and a low-entropy, RSVP-compatible reference:

$$R_S(a, z, K) = S(a, z, K) - S_{\text{ref}}(\mu)(a, z, K),$$

where S_{ref} could be, for instance, the negative log-density or some coarse-grained entropy functional.

5. Combined Equilibrium Operator

Collecting residuals:

$$T(\Phi, \mathbf{v}, S) = \begin{pmatrix} R_V \\ R_\mathbf{v} \\ R_\mu \\ R_K \\ R_S \end{pmatrix}.$$

An Aiyagari equilibrium in RSVP form is then:

$$T(\Phi, \mathbf{v}, S) = 0.$$

D. Specializing the RSVP–Futarchy Lagrangian

We now insert this $T(\Phi, \mathbf{v}, S)$ into the futarchy Lagrangian:

$$\mathcal{L} = \mathcal{L}_{\text{dyn}} + \mathcal{L}_{\text{equil}} + \mathcal{L}_{\text{belief}} + \mathcal{L}_{\text{futarchy}} + \mathcal{L}_{\text{entropy}}.$$

1. Equilibrium Term

The equilibrium constraint becomes:

$$\mathcal{L}_{\text{equil}} = -\frac{\lambda}{2} (|R_V|^2 + \|R_{\mathbf{v}}\|^2 + |R_{\mu}|^2 + |R_K|^2 + |R_S|^2),$$

integrated over (a, z, K) and time.

2. Futarchy & Belief Terms in Aiyagari Context

In this model, the belief field B can be interpreted as a prediction market on future aggregate quantities (e.g. K' , r' , w') and associated welfare. For concreteness, let B encode beliefs about the expected value of some welfare functional $W[\Phi, \mu]$, such as aggregate utility or consumption.

The belief Lagrangian remains:

$$\mathcal{L}_{\text{belief}} = -\frac{\alpha}{2} \|\nabla B\|^2 - \beta B S,$$

interpreting ∇ as gradients in (a, z, K) .

The futarchy coupling term now aligns policy with value gradients in the Aiyagari state space:

$$\mathcal{L}_{\text{futarchy}} = \gamma B \langle \nabla \Phi, \tilde{\mathbf{v}} \rangle - \frac{\kappa}{2} \|\mathbf{v} - \mathbf{v}_B\|^2.$$

Here:

- $\nabla \Phi$ is the gradient of the value function in (a, z, K) ,
- $\tilde{\mathbf{v}}$ is an embedding of the policy vector into the tangent space of $\mathcal{X}$,
- $\mathbf{v}_B$ is the policy implied by the belief field B (e.g. the policy that maximizes predicted aggregate welfare).

Economically, this says:

Policies are driven to align with the value gradient in state space, as weighted by market beliefs, and penalized for deviating from the belief-implied optimal policies.

3. Entropy Term for the Wealth Distribution

The entropy term becomes:

$$\mathcal{L}_{\text{entropy}} = -\eta|\nabla S|^2 - \xi S \log S,$$

where S is related to the cross-sectional density μ (e.g. $S = -\log \mu$ or a smoothed version thereof). This penalizes highly fragmented or unstable distributions and favors smooth, coherent wealth distributions consistent with RSVP lamphrodynamics.

E. Interpretation in the Aiyagari Context

In this concrete setting, the RSVP–Futarchy theory tells the following story:

- $\Phi(a, z, K)$: the household value function and, more broadly, a welfare potential over the microstate space.
- $\mathbf{v}(a, z, K)$: policy rules mapping from states to controls (c, a') .
- $S(a, z, K)$: an entropy field encoding the dispersion and complexity of the wealth and shock distribution.
- $B(a, z, K)$: a predictive field encoding market beliefs about how different states and policies affect future aggregates and welfare.

The equilibrium operator $T(\Phi, \mathbf{v}, S)$ enforces:

1. Bellman consistency,
2. policy optimality,
3. distributional stationarity,
4. market clearing,
5. entropy coherence.

The futarchy coupling terms ensure that:

1. beliefs B aggregate dispersed information about how $(\Phi, \mathbf{v}, S)$ shape future states;
2. policy fields $\mathbf{v}$ are driven toward those that maximize predicted welfare under B ;
3. the resulting dynamics form a lamphrodynamic flow on the space of Aiyagari equilibria, selecting stable, low-entropy, belief-consistent configurations.

Thus a classical incomplete-markets Aiyagari model is realized as a particular instance of the RSVP–Futarchy field theory, with the standard equilibrium conditions emerging as the zero-gradient limit of a more general lamphrodynamic and variational structure.

Appendix: Stationary RSVP Free-Energy for an Aiyagari Equilibrium

In this appendix we specialize the RSVP–Futarchy formulation of the Aiyagari economy to a stationary equilibrium and construct an explicit free-energy functional over the state space. The goal is to make concrete the notion that a stationary Aiyagari equilibrium corresponds to a minimizer of a lamphrodynamic free energy.

A. Stationary Setup

We restrict attention to time-invariant configurations:

$$\partial_t \Phi = 0, \quad \partial_t \mathbf{v} = 0, \quad \partial_t S = 0, \quad \partial_t B = 0.$$

The state space is

$$x = (a, z, K), \quad \mathcal{X} = [\underline{a}, \bar{a}] \times \mathcal{Z} \times [0, \bar{K}],$$

and we assume that the aggregate capital stock is constant at equilibrium, $K = K^*$, so that effectively the state is (a, z) and K^* enters as a parameter.

We write the fields as:

$$\Phi(a, z), \quad \mathbf{v}(a, z) = \begin{pmatrix} c(a, z) \\ a'(a, z) \end{pmatrix}, \quad S(a, z), \quad B(a, z),$$

with K^* implicitly determined by the stationary wealth distribution.

B. Stationary Equilibrium Operator

The equilibrium operator $T(\Phi, \mathbf{v}, S)$ defined previously simplifies under stationarity and constant K^* .

1. Bellman Residual

The Bellman operator becomes:

$$(\mathcal{T}\Phi)(a, z) = \max_{a'} \left\{ u((1 + r^*)a + w^*z - a') + \beta \mathbb{E}[\Phi(a', z') \mid z] \right\},$$

where $r^* = r(K^*)$ and $w^* = w(K^*)$.

The value residual is:

$$R_V(a, z) = \Phi(a, z) - (\mathcal{T}\Phi)(a, z).$$

2. Policy Residual

Define

$$a'_\Phi(a, z) = \arg \max_{a'} \left\{ u((1 + r^*)a + w^*z - a') + \beta \mathbb{E}[\Phi(a', z') \mid z] \right\}.$$

The policy residual remains:

$$R_{\mathbf{v}}(a, z) = \begin{pmatrix} c(a, z) - ((1 + r^*)a + w^*z - a'(a, z)) \\ a'(a, z) - a'_\Phi(a, z) \end{pmatrix}.$$

3. Distribution and Market-Clearing Residuals

Let $\mu(a, z)$ denote the stationary distribution. The law-of-motion residual becomes:

$$R_\mu(a, z) = \mu(a, z) - \mathcal{P}_{\mathbf{v}}[\mu](a, z),$$

where $\mathcal{P}_{\mathbf{v}}[\mu]$ is the stationary fixed point of the Markov operator induced by $(a'(a, z), z')$.

The capital residual simplifies to:

$$R_K = K^* - \int a \, d\mu(a, z).$$

4. Entropy Residual

We define a reference entropy $S_{\text{ref}}(\mu)$, for example

$$S_{\text{ref}}(a, z) = -\log \mu(a, z)$$

(up to normalization). The entropy residual is:

$$R_S(a, z) = S(a, z) - S_{\text{ref}}(a, z).$$

5. Combined Stationary Operator

Thus the stationary equilibrium operator is:

$$T(\Phi, \mathbf{v}, S) = \begin{pmatrix} R_V(a, z) \\ R_{\mathbf{v}}(a, z) \\ R_\mu(a, z) \\ R_K \\ R_S(a, z) \end{pmatrix},$$

and a stationary equilibrium is a zero of T .

C. Stationary Free-Energy Functional

We now define the stationary RSVP free-energy functional:

$$\mathcal{E}[\Phi, \mathbf{v}, S; \mu, K^*] = \frac{1}{2} \int_{\underline{a}}^{\bar{a}} \sum_{z \in \mathcal{Z}} w(a, z) \left(|R_V(a, z)|^2 + \|R_{\mathbf{v}}(a, z)\|^2 + |R_{\mu}(a, z)|^2 + |R_S(a, z)|^2 \right) da + \frac{\omega}{2} |R_K|^2, \quad (61)$$

where $w(a, z)$ is a weighting function (e.g. the stationary density $\mu(a, z)$) and $\omega > 0$ is a weight on the capital residual.

Theorem 6 (Stationary Aiyagari Equilibrium as Free-Energy Minimizer). *Suppose the functional $\mathcal{E}$ is differentiable and the equilibrium operator T is nondegenerate in the sense described in the main text. Then:*

$$(\Phi^*, \mathbf{v}^*, S^*, \mu^*, K^*) \text{ is a stationary Aiyagari equilibrium} \iff (\Phi^*, \mathbf{v}^*, S^*, \mu^*, K^*) \text{ minimizes } \mathcal{E}.$$

Proof. If $T(\Phi^*, \mathbf{v}^*, S^*) = 0$ and $R_K = 0$, then each residual vanishes, implying $\mathcal{E} = 0$, the global minimum. Conversely, if $(\Phi^*, \mathbf{v}^*, S^*, \mu^*, K^*)$ minimizes $\mathcal{E}$, the first variation must vanish. Under the nondegeneracy assumption, this implies each residual is zero, i.e. $T = 0$ and $R_K = 0$. $\square$

This makes precise the statement that the stationary equilibrium of an Aiyagari economy is a lamphrodynamically relaxed configuration in RSVP field space.

D. Stationary Lamphrodynamic Gradient Flow

Even in the stationary setting, one can define an abstract gradient flow in a fictitious “relaxation time” τ :

$$\frac{\partial \Phi}{\partial \tau} = -\frac{\delta \mathcal{E}}{\delta \Phi}, \quad (62)$$

$$\frac{\partial \mathbf{v}}{\partial \tau} = -\frac{\delta \mathcal{E}}{\delta \mathbf{v}}, \quad (63)$$

$$\frac{\partial S}{\partial \tau} = -\frac{\delta \mathcal{E}}{\delta S}, \quad (64)$$

$$\frac{\partial \mu}{\partial \tau} = -\frac{\delta \mathcal{E}}{\delta \mu}, \quad (65)$$

$$\frac{\partial K^*}{\partial \tau} = -\frac{\partial \mathcal{E}}{\partial K^*}. \quad (66)$$

This flow monotonically decreases $\mathcal{E}$ and converges, under suitable conditions, to the stationary equilibrium. In numerical deep learning solvers, discrete steps of stochastic gradient descent on network parameters approximate this continuum lamphrodynamic flow.

Appendix: RSVP–Futarchy Formulation of a HANK-Style New Keynesian Economy

In this appendix we sketch a Heterogeneous-Agent New Keynesian (HANK) economy in RSVP–Futarchy terms. The goal is to show how macro policy rules (monetary and/or fiscal), shocks, and heterogeneous households fit naturally into the $(\Phi, \mathbf{v}, S, B)$ field-theoretic framework.

A. Stylized HANK New Keynesian Setup

We consider a stylized HANK environment with:

- heterogeneous households indexed by assets a , income states z ;
- aggregate variables: output gap x_t , inflation π_t , nominal interest rate i_t ;
- a monetary policy rule and possibly fiscal instruments;
- idiosyncratic and aggregate shocks.

For concreteness, suppose aggregate dynamics obey (in log-linear form):

$$x_t = \mathbb{E}_t x_{t+1} - \sigma(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n), \quad (67)$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t, \quad (68)$$

where u_t is a cost-push shock and r_t^n the natural real rate.

A Taylor-type monetary rule:

$$i_t = \bar{i} + \phi_\pi \pi_t + \phi_x x_t + \varepsilon_t,$$

with ε_t a monetary policy shock.

Heterogeneous households face incomplete markets and solve problems similar in spirit to the Aiyagari households, but in a time-varying aggregate environment.

B. State Space and RSVP Fields in HANK–NK

We define a combined micro–macro state:

$$x = (a, z, x^A), \quad x^A = (x, \pi, i, \dots),$$

where x^A collects aggregate variables (output gap, inflation, rate, and possibly others).

The state manifold is:

$$\mathcal{X} = [a_{\min}, a_{\max}] \times \mathcal{Z} \times \mathcal{X}^A,$$

where $\mathcal{X}^A$ is the macro-state space.

We define RSVP fields as:

- **Value Field** $\Phi(a, z, x^A)$: HANK value function, incorporating both micro and macro states.
- **Policy Field** $\mathbf{v}(a, z, x^A)$: a vector of household controls and possibly macro policy levers. For instance:

$$\mathbf{v} = \begin{pmatrix} c(a, z, x^A) \\ a'(a, z, x^A) \end{pmatrix} \quad (\text{micro}),$$

plus

$$\mathbf{v}^{\text{pol}}(x^A) = \begin{pmatrix} i(x^A) \\ f(x^A) \end{pmatrix} \quad (\text{macro policy rule: monetary } i, \text{ fiscal } f).$$

For brevity, we write the total policy field as $\mathbf{v}$, with micro and macro components.

- **Entropy Field** $S(a, z, x^A)$: encoding the complexity of the joint distribution over assets, idiosyncratic states, and macro configurations.
- **Belief Field** $B(a, z, x^A)$: a futarchy belief field representing prediction-market beliefs over future aggregates and welfare outcomes given candidate policy rules.

C. Equilibrium Operator in HANK–NK

We sketch the main components of $T(\Phi, \mathbf{v}, S)$.

1. Micro Bellman Residual

Households solve:

$$\Phi(a, z, x^A) = \max_{a'} \left\{ u(c(a, z, x^A)) + \beta \mathbb{E}[\Phi(a', z', x^{A'}) \mid a, z, x^A] \right\},$$

where c obeys budget constraints given prices determined by $x^A = (x, \pi, i, \dots)$ and policy $\mathbf{v}^{\text{pol}}$.

We define:

$$R_V(a, z, x^A) = \Phi(a, z, x^A) - (\mathcal{T}\Phi)(a, z, x^A),$$

where $\mathcal{T}$ is the corresponding Bellman operator.

2. Micro Policy Residual

As in the Aiyagari case, define an optimal a'_Φ and construct:

$$R_{\mathbf{v}}^{\text{micro}}(a, z, x^A) = \begin{pmatrix} c(a, z, x^A) - \text{budget-consistent}(a, z, x^A, a') \\ a'(a, z, x^A) - a'_\Phi(a, z, x^A) \end{pmatrix}.$$

3. Macro NK Residuals

We define macro residuals for the NK block:

$$R_x(x^A) = x - [\mathbb{E}x' - \sigma(i - \mathbb{E}\pi' - r^n)], \quad (69)$$

$$R_\pi(x^A) = \pi - [\beta\mathbb{E}\pi' + \kappa x + u], \quad (70)$$

$$R_i(x^A) = i - [\bar{i} + \phi_\pi\pi + \phi_x x + \varepsilon]. \quad (71)$$

These can be collected in a macro residual vector:

$$R_{\text{NK}}(x^A) = \begin{pmatrix} R_x(x^A) \\ R_\pi(x^A) \\ R_i(x^A) \end{pmatrix}.$$

4. Distribution and Entropy Residuals

Let $\mu(a, z, x^A)$ be the joint distribution over micro and macro states (or a conditional distribution over (a, z) given x^A and an aggregate law of motion for x^A). We define:

$$R_\mu(a, z, x^A) = \mu(a, z, x^A) - \mathcal{P}_{\mathbf{v}}[\mu](a, z, x^A),$$

and a reference entropy $S_{\text{ref}}(\mu)$, with:

$$R_S(a, z, x^A) = S(a, z, x^A) - S_{\text{ref}}(a, z, x^A).$$

5. Combined HANK–NK Operator

Collecting all components:

$$T(\Phi, \mathbf{v}, S) = \begin{pmatrix} R_V(a, z, x^A) \\ R_{\mathbf{v}}^{\text{micro}}(a, z, x^A) \\ R_{\text{NK}}(x^A) \\ R_\mu(a, z, x^A) \\ R_S(a, z, x^A) \end{pmatrix}.$$

A HANK–NK equilibrium satisfies $T(\Phi, \mathbf{v}, S) = 0$.

D. Futarchy Coupling for Macro Policy Rules

In a futarchy setting, prediction markets bet on the consequences of alternative macro policy rules. Let θ denote rule coefficients (e.g. ϕ_π, ϕ_x , or parameters of a richer monetary/fiscal rule), and encode them as part of the macro policy field $\mathbf{v}^{\text{pol}}$.

The futarchy term in the Lagrangian is adapted to:

$$\mathcal{L}_{\text{futarchy}}^{\text{HANK-NK}} = \gamma B(a, z, x^A) \langle \nabla \Phi(a, z, x^A), \tilde{\mathbf{v}}(a, z, x^A) \rangle - \frac{\kappa}{2} \|\mathbf{v}^{\text{pol}}(x^A) - \mathbf{v}_B^{\text{pol}}(x^A)\|^2.$$

Here:

- B aggregates predictions over future welfare given policy rule parameters θ ,
- $\mathbf{v}_B^{\text{pol}}$ is the policy rule that maximizes expected welfare under beliefs B ,
- $\tilde{\mathbf{v}}$ is the embedding of micro/macro policy directions into the tangent space of $\mathcal{X}$.

This implements Hansons principle:

choose policy rules that maximize predicted welfare.

E. HANK–NK Free-Energy Functional

We can now define a HANK–NK RSVP free-energy functional:

$$\mathcal{E}_{\text{HANK-NK}}[\Phi, \mathbf{v}, S] = \frac{1}{2} \int_{\mathcal{X}} \left(|R_V|^2 + \|R_{\mathbf{v}}^{\text{micro}}\|^2 + |R_\mu|^2 + |R_S|^2 \right) d\mu_x \quad (72)$$

$$+ \frac{1}{2} \int_{\mathcal{X}^A} \|R_{\text{NK}}(x^A)\|^2 d\mu_A, \quad (73)$$

where $d\mu_x$ and $d\mu_A$ are appropriate measures over the full state space and macro space, respectively.

Theorem 7 (HANK–NK Equilibrium as RSVP Free-Energy Minimizer). *Under regularity and nondegeneracy assumptions analogous to those in the stationary Aiyagari case,*

$$T(\Phi^*, \mathbf{v}^*, S^*) = 0 \quad \Longleftrightarrow \quad (\Phi^*, \mathbf{v}^*, S^*) \text{ minimizes } \mathcal{E}_{\text{HANK-NK}}.$$

Proof. Identical in structure to the Aiyagari free-energy theorem: if all residuals vanish, $\mathcal{E}_{\text{HANK-NK}} = 0$, the global minimum. Conversely, a minimizer must have vanishing first variation, which implies each residual is zero under nondegeneracy. $\square$

Thus a HANK–NK equilibrium emerges as a lamphrodynamically relaxed configuration of the RSVP fields, with futarchy couplings directing the macro policy rule toward belief-weighted welfare optima.

F. Interpretation

In summary:

- The *micro* side (heterogeneous households) is captured by $\Phi(a, z, x^A)$, $\mathbf{v}^{\text{micro}}$, and $S(a, z, x^A)$.
- The *macro* side (NK block and policy rules) is captured by $\mathbf{v}^{\text{pol}}(x^A)$, $R_{\text{NK}}(x^A)$, and their contribution to the free energy.
- The *belief* field $B(a, z, x^A)$ integrates predictive information about the macro effects of policy rules, and, once integrated out, induces nonlocal couplings in the effective action.
- Equilibrium is the joint minimum of the free-energy functional over both micro and macro fields, respecting futarchy's separation of values (encoded in Φ) and beliefs (encoded in B).

This closes the loop between HANK microfoundations, New Keynesian aggregate dynamics, deep learning-style lamphrodynamic solvers, and Hanson-style futarchy within a single RSVP field-theoretic framework.